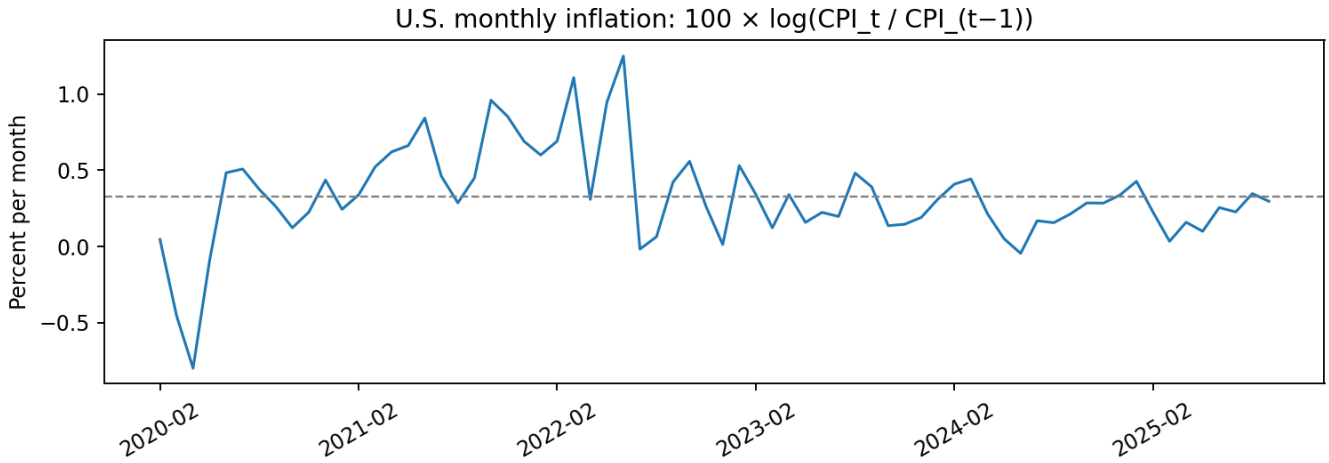
WEEK 5 | TIME-SERIES COMPUTATIONAL EXERCISE

U.S. monthly inflation: AR model selection and stationarity

DATA & METHOD

Source: FRED / BLS, seasonally adjusted CPI (CPIAUCSL), Jan 2020–Sep 2025. Monthly inflation = 100 × ln(CPI_t / CPI_(t-1)). I fit AR(1) to AR(6) using a common 62-month estimation sample (Aug 2020–Sep 2025).

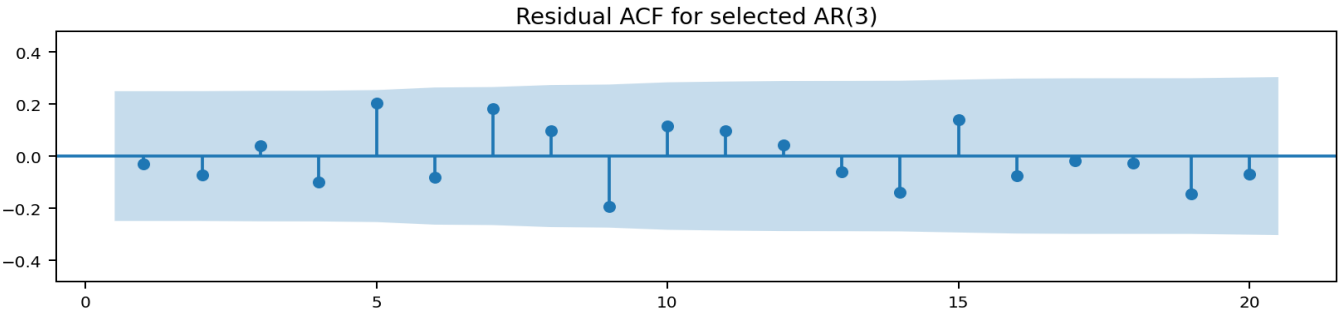


MODEL COMPARISON

Model	AIC ↓	BIC ↓	Ljung–Box p (12 lags)
AR(1)	−173.36	−169.11	0.006
AR(2)	−171.40	−165.02	0.004
AR(3)	**−179.03**	**−170.52**	**0.194**
AR(4)	−177.07	−166.44	0.151
AR(5)	−176.72	−163.96	0.099
AR(6)	−174.77	−159.88	0.073

Selected AR(3): inflation_t = 0.1254 + 0.4858 inflation_(t−1) − 0.1963 inflation_(t−2) + 0.3664 inflation_(t−3)

RESIDUAL CHECK



Ljung–Box p = 0.194 at lag 12 (> 0.05): fail to reject no residual autocorrelation across the tested lags. The residual ACF shows no large remaining individual autocorrelations.

ADF UNIT-ROOT TESTS

Series	ADF statistic	p-value
Monthly inflation (level)	−4.031	0.0013
First difference of inflation	−6.295	< 0.001

Interpretation: both tests reject the unit-root null at 5%. Monthly inflation already appears I(0) over this sample, so a first difference is not required to remove a unit root. This does not imply that the CPI price level itself is stationary.

TAKEAWAY

AR(3) provides the lowest AIC and BIC among the six fitted models. The Ljung–Box check does not find strong leftover serial correlation, and the ADF results support stationary monthly inflation for this sample. The 2020–2022 inflation episode and the short sample limit how far this result should be generalized.

Data: https://fred.stlouisfed.org/series/CPIAUCSL

AIC/BIC are comparable Gaussian OLS criteria with shared likelihood constants omitted.